

CAPSTONE PROJECT REPORT

(Project Term January-May 2026)

Smart Drug Inventory & Expiry Tracking System for Small Clinics / Pharmacies

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Course Code CSE439

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P ROFESSIONAL
U NIVERSITY

PAC Form



TOPIC APPROVAL PERFORMA

School of Computer Science and Engineering

Program : P132::B.Tech. (Computer Science and Engineering)

COURSE CODE : CSE439

REGULAR/BACKLOG : Regular

GROUP NUMBER : 2RGC0246

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PROPOSED TOPIC : Smart Drug Inventory & Expiry Tracking System for Small Clinics / Pharmacies

Qualitative Assessment of Proposed Topic by PAC		
Sr.No.	Parameter	Rating (out of 10)
1	Project Novelty: Potential of the project to create new knowledge	7.50
2	Project Feasibility: Project can be timely carried out in-house with low-cost and available resources in the University by the students.	7.83
3	Project Academic Inputs: Project topic is relevant and makes extensive use of academic inputs in UG program and serves as a culminating effort for core study area of the degree program.	7.50
4	Project Supervision: Project supervisor's is technically competent to guide students, resolve any issues, and impart necessary skills.	8.00
5	Social Applicability: Project work intends to solve a practical problem.	8.00
6	Future Scope: Project has potential to become basis of future research work, publication or patent.	7.50

PAC Committee Members		
PAC Member (HOD/Chairperson) Name: Dr. Mohit Arora	UID: 15980	Recommended (Y/N): Yes
PAC Member (Allied) Name: Dr. Ankita Wadhawan	UID: 23891	Recommended (Y/N): Yes
PAC Member 3 Name: Upinder Kaur	UID: 27954	Recommended (Y/N): Yes

Final Topic Approved by PAC: Smart Drug Inventory & Expiry Tracking System for Small Clinics / Pharmacies

Overall Remarks: Approved

PAC CHAIRPERSON Name: 11265::Dr. Dalwinder Singh

Approval Date: 05 Dec 2025

DECLARATION

We hereby declare that the project work entitled “**Smart Drug Inventory & Expiry Tracking System for Small Clinics / Pharmacies**” is an authentic record of our own work carried out as requirements of Capstone Project for the award of B.Tech degree in Computer Science and Engineering from Lovely Professional University, Phagwara, under the guidance of (Name of Faculty Mentor), during August to November 2022. All the information furnished in this capstone project report is based on our own intensive work and is genuine.

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CERTIFICATE

This is to certify that the declaration statement made by this group of students is correct to the best of my knowledge and belief. They have completed this Capstone Project under my guidance and supervision. The present work is the result of their original investigation, effort and study. No part of the work has ever been submitted for any other degree at any University. The Capstone Project is fit for the submission and partial fulfillment of the conditions for the award of B.Tech degree in Computer Science Engineering from Lovely Professional University, Phagwara.

Signature and Name of the Mentor

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We humbly take this opportunity to present our votes of thanks to all those guideposts who really acted as lightening pillars to enlighten our way throughout this project that has led to successful and satisfactory completion of this study.

We are grateful to our mentor Vishal Khanna for providing us with an opportunity to undertake this project in this university and providing us with all the bright and innovative ideas for making our project a worthwhile of running in an organization. We are highly thankful for his active support, valuable time and advice, whole-hearted guidance, sincere cooperation, and pains-taking involvement during the study and in completing the capstone project within the time stipulated.

We are thankful to all those, particularly our friends, who have been instrumental in creating proper, healthy, and conductive environment and including new and fresh innovative ideas for us during the project, without their help, it would have been extremely difficult for us to prepare the project in a time bound framework.

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Bachelor of Technology
(Computer Science and Engineering)

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CHAPTER 1: INTRODUCTION

Healthcare management at the grassroots level in India faces many unique challenges which affect the quality of care given to patients. For instance, the primary healthcare system in India relies heavily on small clinics and pharmacies; however, these small businesses frequently find themselves managing their pharmaceutical inventory ineffectively since many of them have hundreds of different products with many different batch numbers, suppliers, quantity levels, and expiration dates of products but do not have any type of secure digital system in place to help keep track of these items therefore resulting in two recurring and easily preventable crises; one is wasting medication due to old stock sitting on their shelves that they are unable to find right away because they do not have a way to track everything adequately and secondly, having an unexpected stockout of critical care medications for a patient when they most need it.

In the current situation, the manual system that uses paper registers, handwritten ledgers, and simple spreadsheet-based systems has significant limitations regarding their reliability. Errors in entry have not been corrected; expiry dates have been missed and there is no means of providing proactive alerts to manage potentially expired medicines. Pharmacists and staff reviewing the medicines for the pharmacy or clinic must continually search through every item to determine if they will be expiring soon; this is not only labor intensive, but also prone to error. The refill process is reactive as stock levels are only checked after being depleted due to stockout; therefore, orders placed subsequent to the last refill will cause delays in filling the prescription and increase the cost of the product. As a result of this process's inefficiencies, there is considerable waste of financial resources, harm to patients, and increased administrative workload for an already overloaded workforce in a health system.

The Smart Drug Inventory/Expiration tracking system was designed to address this issue by creating a mobile-focused, intelligent platform (health technology) that enhances

Digital Inventory Management in small clinics and pharmacies, with the goal of improving disciplined, efficient digital

Management of Supplies. The system has two interconnected parts: an android mobile application (main operational user interface for pharmacy/clinic staff) and a web-based solution providing supplemental product information (ie. item details, app download, AI-powered

chatbot, feedback mechanism) to support/prioritize use of the smart drug inventory/expiry tracking system.

The project is a direct outcome of the Smart India Hackathon 2024 (SIH) problem statement SIH1627 issued by the Government of India to address the larger problem of drug inventory and supply chain tracking across healthcare facilities across the country. The project will tackle the problem at the level of the smallest healthcare units – single-pharmacy clinics and small dispensaries – to develop a scalable, affordable and accessible solution that can be incorporated into national drug supply systems.

1.1 BACKGROUND AND MOTIVATION

India has over a million registered pharmacies and hundreds of thousands of small clinics across urban, semi-urban and rural areas. The majority of these facilities do not have a formal digital inventory management system. A 2022 study by the Indian Pharmaceutical Alliance estimated that almost 4 to 5 percent of medicines dispensed in India are either expired or near-expiry at the time of dispensing, representing both a patient safety risk and a significant economic loss. “Small clinics operate on very slim margins and wastage of medicines can have a serious impact on finances,” said Dr.

As of 2024, India has seen an unparalleled penetration of Android smartphones with more than 600 million active Android users. There is an opportunity, due to the availability of cheap Android devices on a large scale. An Android based inventory management application can be deployed to almost any pharmacy in India, without any investment in hardware, other than the smartphone which the pharmacist already owns. Taking advantage of this opportunity, the Smart Drug Inventory and Expiry Tracking System provides a fully functional Android application that works offline, does not need dedicated hardware, does not require an enterprise software subscription, and does not need technical expertise to run.

1.2 OBJECTIVES OF THE PROJECT

The following is a list of primary goals for this project:

1. To develop an entirely functional Android mobile application providing digital drug inventory management for pharmacists to add and update record details including batch number, quantity, category, manufacturer, unit price, and expiration date, as well as track each medicine.

2. To implement an intelligent automated expiry tracking system that monitors all medicines and generates tiered alerts at 30 days, 15 days, and 7 days before a medicine's expiry date, providing adequate time for corrective action.
3. To develop a proactive stock monitoring system with configurable minimum quantity thresholds per medicine to create alerts prior to a stockout.
4. To integrate a real-time push notification system using Firebase Cloud Messaging to deliver expiration and low stock alerts to the pharmacist's mobile device regardless of whether the mobile app is actively being used to manage drug inventory.
5. To create a data-driven analytics dashboard that provides visual insights including the historical usage trend by month, the distribution of drug categories in the overall inventory, total value of the entire inventory, and a comparison of expired-to-active drug stock.
6. To create a vendor and purchase management module, creating a single data repository for all supplier contact information, purchase order history, and transaction history within a single application.
7. To develop a supporting web platform, which provides product information, APK download, AI-powered chatbot to assist users, and mechanism for contact and feedback.
8. Align the system with the United Nations Sustainable Development Goals: SDG 3 (Good Health and Well-being), SDG 9 (Industry, Innovation and Infrastructure), SDG 12 (Responsible Consumption and Production) and SDG 11 (Sustainable Cities and Communities).

1.3 SCOPE OF THE PROJECT

This is a Capstone Project developed in two semesters. Capstone Project-1 (CP-1) is the complete system design, requirements specification and development of the core Android application with all the primary modules and the web platform. The Capstone Project-2 (CP-2) will improve the system by adding cloud synchronization, barcode scanning, multi-store management, AI-based demand prediction and will deploy the system to Google Play Store.

The system is targeted for use in small to medium sized pharmacies and single doctor clinics in semi-urban and urban India. Phase 1: The target user is a single pharmacist or clinic staff member at a single pharmacy location, using an Android smartphone with Android 8.0 or above.

1.4 REPORT ORGANIZATION

The format of this report is as follows; Chapter 2 Provides an overview of the problem and details regarding the scope of the study. Chapter 3 discusses the current system as well as limitations associated with that system. Chapter 4 contains an analysis of the problem including feasibility and project planning. Chapter 5 Contains a complete software requirements analysis. Chapter 6 Contains system design as well as data flow diagrams, entity relationship diagrams and database schemas. Chapter 7 Contains a description of the overall test strategy. Chapter 8 Describes how systems will be implemented. Chapter 9 Describes project legacy and lessons learned. Chapter 10 Contains the user guide for projects. Chapter 11 Contains the various snapshots of the finished product. Finally, chapter 12 Contains the bibliography.

CHAPTER 2: PROFILE OF THE PROBLEM – RATIONALE AND SCOPE OF THE STUDY

2.1 PROBLEM STATEMENT

Pharmacies and small clinics in India don't have easy-to-access and affordable digital solutions to manage their pharmaceutical inventories. The use of a manual register, handwritten stock cards and simple spreadsheets results in the following issues impacting a pharmacy's or clinic's ability to function well: (i) undetected medicine expired, (ii) unanticipated stock shortages of critical or essential medications, (iii) incorrect records of medicine purchased, and (iv) poor relationships with vendors — all of which have an immediate and measurable impact on patient safety, quality healthcare services, and the overall financial sustainability of small health facilities.

The problem that this project is attempting to address can be formally defined as follows: what are the design and development (for implementation) requirements for a mobile-based and reasonable-priced digital solution to help pharmacists and staff of small health facilities proactively manage their drug inventories, monitor medicine expiration dates with the use of automated notifications, and obtain relevant and actionable analytics concerning prescription and over-the-counter (OTC) medication (without the necessity of any special equipment, internet connectivity for the core features, or technical expertise)?

2.2 RATIONALE OF THE STUDY

The rationale for commencing this project is based on four intersecting elements:

The magnitude of the issue is quite large. India has about 1 million pharmacies, most of which lack digital inventory systems. Even small increases in the detection of expired items as well as the management of inventory can lead to a large decrease in the number of times that expired items are dispensed and save the pharmaceutical industry crores of rupees per year due to the amount of medicines that will be unusable or wasted.

The target users have access to the technology necessary to address this issue. Most pharmacies in India use Android smartphones. Well-designed Android apps add no additional cost for hardware and can be installed in a few minutes. The offline capability of Room Database means

that after downloading the app the essential functions will work even without an Internet connection. This addresses the concerns that are normally associated with a poor quality network.

The existing solutions are too expensive for small pharmacies, very complicated to use due to a lack of technical skills, or designed for large enterprise companies that do not reflect the physical makeup of a small clinic. The market has a significant void for a low-cost, simple solution specifically designed for small health care providers.

Fourth, this challenge is congruent with nationwide goals established by the Government of India (GoI). Specifically, SIH1627 in the Smart India Hackathon 2024 (SIH) contest identifies drug inventory/supply chain traceability among priority areas for technology-based solutions. As a group, the project is a contribution to an identified problem area that has been deemed to be of national importance by the Government of India.

2.3 SCOPE OF THE STUDY

2.3.1 In Scope

- An Android app that consists of 7 core modules: user authentication; inventory management; item expiry tracking; monitoring item stock, push notifications; analytics dashboard; vendor & purchase management.
- A web platform that includes a landing page, APK download function, AI-powered chatbot, & a contact form.
- A local database that allows for offline use of the app's data. The database is stored in CP-1.
- Push notifications will be served via Firebase Cloud Messaging.
- System documentation including DFDs, ER diagram, database schema, UI wireframes.
- A complete test plan will cover functional testing, structural testing, & integration testing.

2.3.2 Out of Scope (Planned for CP-2)

- Scanning of barcodes & QR codes as a method of automatically entering the medicine information into the database.

- Support for multiple stores & numerous branches.
- Synchronization between devices using cloud computing.
- Integrate with governmental organisations that regulate the supply of drugs throughout the country.
- Use artificial intelligence to predict demand based on historical usages & provide suggestion(s) for automatic reordering of items.
- Ability to export client reports to either an Excel file or a PDF format.
- Multi-lingual application support (English/Hindi and a broad range of regional languages).

2.4 COMPARISON BETWEEN EXISTING AND PROPOSED SYSTEM

Table 2.1: Comparison Between Existing and Proposed System

Feature	Existing System	Proposed System
Data Management	Manual / Paper-based	Digital – Android App
Expiry Tracking	Manual inspection only	Automated – 30/15/7 day alerts
Stock Monitoring	Reactive – checked after stockout	Proactive – configurable threshold alerts
Push Notifications	Not Available	Firebase Cloud Messaging – real-time
Analytics Dashboard	Not Available	Visual charts – usage trends, pie charts
Vendor Management	Separate registers	Integrated within the same app
Data Accuracy	Error-prone manual entry	Validated form-based digital entry
User Accessibility	Requires physical presence	Mobile – accessible anytime, anywhere
Cost	Low (paper) but high wastage cost	Near-zero software cost; free tier cloud
Scalability	Not scalable	Cloud-ready architecture for multi-clinic in CP-2

2.5 TARGET USERS AND BENEFICIARIES

The main people who benefit from this system are pharmacists and clinic staff who work at pharmacies or clinics. These people benefit because they can manage medicines better.

The patients who visit these pharmacies and clinics also benefit. They get medicines that are not expired. They are less likely to get medicines that're about to expire. Healthcare

administrators and government health agencies also benefit in the run. They can use information from these systems to make decisions, about the medicine supply chain. This can help them at an national level. They can make sure that medicines are available where they are needed. The information can help them make choices.

2.6 INTERDISCIPLINARY NATURE OF THE PROJECT

This project is really about bringing different areas of study. It uses ideas from these areas:

- Computer Science and Engineering: we are talking about building Android apps designing databases creating software architecture using REST APIs and sending push notifications.
- Pharmaceutical/Healthcare Administration: Knowledge of drug stocking guidelines and regulations; knowledge of expiration rules; knowledge of drug supply chain requirements; and knowledge of the workflow of a pharmacist.
- Data Analytics: Construction of an analytics dashboard; calculation of usage trends; calculation of inventory statistics.
- Human-Computer Interaction: Designing a user interface that is focused on accessibility for non-technical healthcare users.

2.7 SUSTAINABLE DEVELOPMENT GOALS ALIGNMENT

Table 2.2: SDG Alignment of the Project

SDG Goal	Relevance to This Project
SDG 3 – Good Health and Well-being	Reduces dispensing of expired medicines; ensures medicine availability; improves quality of care at small healthcare facilities.
SDG 9 – Industry, Innovation, and Infrastructure	Introduces digital infrastructure to small clinics that currently lack any technology-based inventory management.
SDG 12 – Responsible Consumption and Production	Reduces pharmaceutical wastage caused by undetected expiry; promotes efficient use of medicine stock.
SDG 11 – Sustainable Cities and Communities	Strengthens healthcare resilience of urban and semi-urban communities by improving medicine availability at local facilities.

CHAPTER 3: EXISTING SYSTEM

3.1 INTRODUCTION TO THE EXISTING SYSTEM

If you want to create an effective replacement, your first step is to identify what systems small clinics/pharmacies currently use to manage drug inventories. The majority of small healthcare facilities in India still use three primary ways to manage their drug inventories - either fully manual with paper-based inventory tracking registers; basic digital tools like Excel spreadsheets; or generic inventory management software that is not designed specifically for conducting pharmacy operations. Each of these systems has several significant limitations that this project hopes to overcome.

3.2 EXISTING SOFTWARE AND MANUAL SYSTEMS

3.2.1 Paper-Based Manual Registers

While Inventory management in many small clinics and pharmacies is typically done manually by keeping an inventory log or ledger where all medications, including the medication name and quantity, as well as the expiration dates, are recorded by hand. Each time a new supply of medications is delivered by a vendor, the additional amount is recorded in the log; conversely, you will notice when a medication is given to a patient by finding it in the log and referencing the reduction of inventory associated with that medication.

The following are the features and drawbacks associated with this approach.

- **Manual Checks of Expiration Dates** — Since there are no automated reminders or alerts for expired medications, the pharmacist is responsible for periodically going through the log of all medications and manually scanning for expired medications.
- **Potential for Data Entry Errors** — There is a high potential for errors in data entry caused by either the use of illegible handwriting, incorrect data entries such as incorrect dates, or missing entries altogether.
- **No Search/Filter Function** — In order to find records for a particular medication, you must look through multiple pages of the log.
- **No Analytics** — To understand trends in usage for a medication, you must manually count the entries for the medication in the log.

- Potential for Physical Damage — The log may become destroyed through being lost, damaged from fire or water, or may simply become illegible over time.
- No Backup — If the log is lost or destroyed, there is no data backup available..

3.2.2 Microsoft Excel Spreadsheets

While some pharmacies may have chosen to utilize simple Excel spreadsheets instead of keeping paper logs of customer transactions, they can only improve searchability and data entry accuracy marginally; they still lack several vital features:

- No automated expiration reminder functionality - Pharmacists will have to manually filter/sort and identify product expiration date.
- No push notifications - The user will have to actively open the spreadsheet in order to check for product.
- No mobile-friendly interface - Excel spreadsheets are extremely difficult to work with via smartphones, which is the only device available to most independent pharmacies.
- No integrated cloud or notification functionality - Product data only exists locally where the information is saved.
- Requires minimal proficiency in Excel. Many independent pharmacies may not be comfortable using Excel.
- No module to manage vendor or purchase orders.

3.2.3 Generic Inventory Management Software

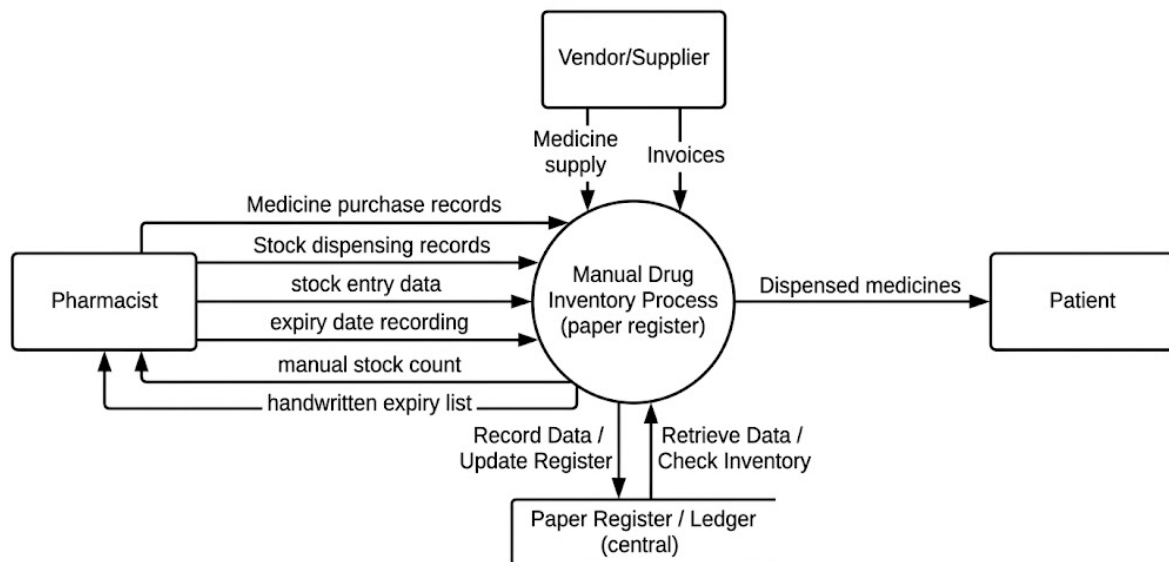
Certain generic retail inventory management applications are available in the market. However, these are typically designed for general retail businesses and lack pharmaceutical-specific features such as batch number tracking, expiry date monitoring, drug category classification, and regulatory compliance support. They are also often priced beyond the budget of small clinics and pharmacies, and require training that is not feasible for small operations.

Table 3.1: Limitations of Existing Systems

Limitation	Paper Register	Excel Spreadsheet	Generic Software
Automated expiry alerts	No	No	Limited
Mobile-native interface	No	No	Sometimes
Push notifications	No	No	No
Offline capability	Yes (paper)	Yes (local file)	Rarely
Vendor management	Manual register	Separate sheet	Sometimes
Analytics dashboard	No	Manual charts	Basic
Pharmaceutical-specific features	No	No	No
Affordable for small clinics	Yes	Yes	Often No
Risk of data loss	High	Medium	Low

3.3 DFD FOR THE PRESENT SYSTEM

The Data Flow Diagram (DFD) for the existing system represents the manual processes currently used to manage drug inventory in a small pharmacy. The present system has no software component — all data flows are paper-based and manual.



Note: In the existing system, all processes are manual with no automated alerts, no digital search, and no notification mechanism. Data exists only in physical form.

Figure 3.1: Level 0 DFD for the Existing Manual Drug Inventory System

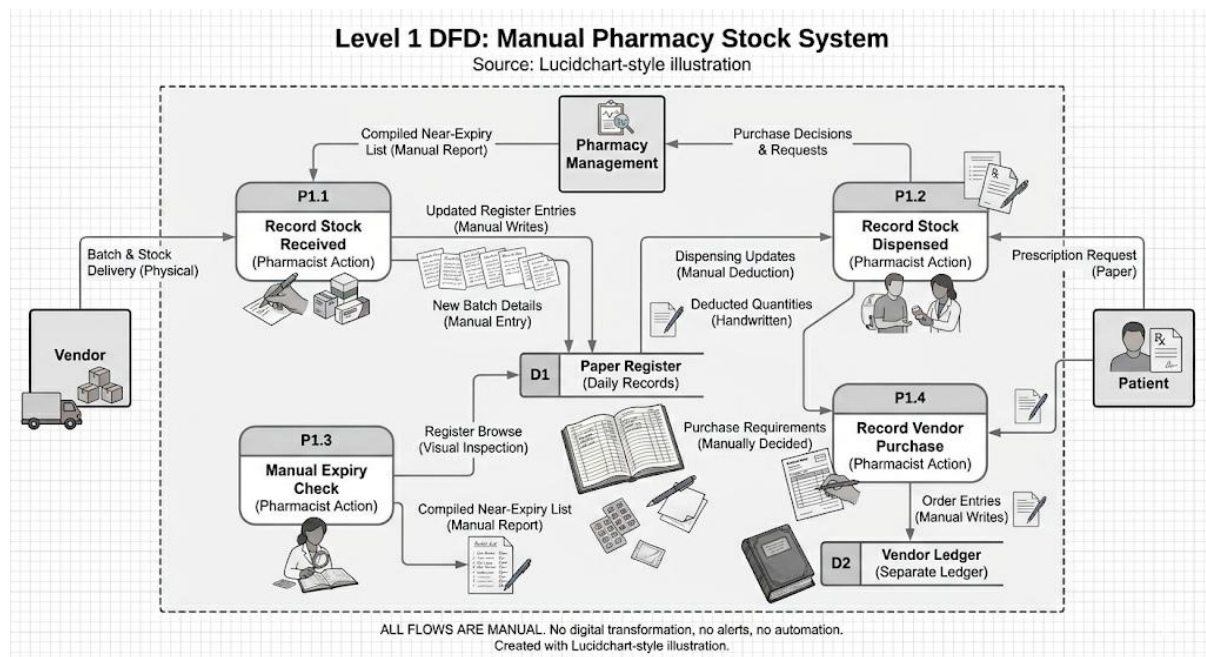


Figure 3.2: Level 1 DFD for the Existing Manual System

3.4 WHAT IS NEW IN THE SYSTEM TO BE DEVELOPED

The Data Flow Diagram (DFD) of the existing system depicts a manual methodology for drug inventory management in a small pharmacy. The existing process is entirely comprised of paper documentation and manual processes with no technological or digital component whatsoever.

The following key features of the proposed Smart Drug Inventory and Expiry Tracking System will improve these manual and semi-digital systems significantly:

- **Automated Expiration Tracking:** This system has a background scheduler that checks all medicine expiration dates on a daily basis and creates tiered alarms (30, 15 and 7 days prior) so that all expired product can be tracked electronically instead of manually.
- **Real-time Push Notifications:** All expiration and low-stock alarms are delivered using Firebase Cloud Messaging to the pharmacist's device, even if the application is not currently running, which will help ensure that no alarm is missed by a pharmacist.
- **Proactive Stock Monitoring:** Each medicine can have its own configurability for establishing minimum stock thresholds and, as such, the pharmacist will receive alerts prior to experiencing stock-outs.
- **Mobile-Native Interface:** The Android application has been developed specifically for use with smartphones with an interface based upon Material Design 3 that is designed for one-handed use in a pharmacy with a busy environment.
- **Offline Capability:** Room Database provides full offline functionality for all core inventory operations — medicines can be added, edited, viewed, and deleted without internet connectivity.

- All the inventory operations that the medicine store does from adding to modifying, viewing to deleting are done completely offline, meaning that you don't need the Internet or mobile data to do any of these operations thanks to the use of database Rooms, so there is no limitation based on connectivity.
- In place of counting inventory on paper or using Excel spreadsheets in order to gather insights into that inventory, charts that provide visuals and at-a-glance summary data now provide real time charts of how that inventory is doing today against yesterday and historical averages.
- Supplier information and purchase orders can now be managed within the same solution as your inventory, which eliminates the need for vendor and vendor order tracking systems on separate spreadsheets.
- An AI-powered chatbot is located on the website to handle instant support requests from users along with frequently asked questions, which means you do not have to provide human support personnel for each of the questions or requests from users.
- Medicine cards (aka inventory items) are colour coded on the inventory list: green for safe, yellow for expiring soon, and red for critical or expired; thus allowing for a visual assessment of the item status (no need to look at A records).
- Firebase Authentication provides security for user accounts, and the architecture supports backup of data as the cloud storage solution is certified to CP-2; thus eliminating the likelihood of data being lost due to the physical destruction of an inventory management register.

CHAPTER 4: PROBLEM ANALYSIS

4.1 PRODUCT DEFINITION

The Smart Drug Inventory and Expiry Tracking Solution consists of two parts:

Mobile Application (Android): This serves as the main operating interface. This component was constructed using Kotlin with an MVVM design using the Room Database for local storage, as well as Firebase for authentication and push notifications. The Mobile Application will allow pharmacists to efficiently manage their drug inventory, including providing an expiry tracking system for managing drugs, keeping track of stocks, gaining analytics on inventory use, and managing their vendors.

Web UI: The web UI will serve as a supporting information and assistance portal. This part will contain a general description of the system, an APK download link for the Mobile Application, a chatbot powered by AI to assist users via the Web UI, and a method for users to contact us and provide feedback.

Table 4.1: Product Definition Summary

Attribute	Details
Product Name	Smart Drug Inventory and Expiry Tracking System
Product Type	Mobile-first Android Application + Supporting Web Platform
Primary Platform	Android 8.0 (API Level 26) and above
Primary Language (App)	Kotlin
Architecture Pattern	MVVM (Model-View-ViewModel) with Repository Pattern
Local Database	Room Database (SQLite abstraction)
Cloud Services	Firebase Authentication + Firebase Cloud Messaging
Web Technology	Standard web stack with Claude API chatbot integration
Distribution Method	APK sideload via website (CP-1); Google Play Store (CP-2)
Inspiration	SIH1627 – Drug Inventory and Supply Chain Tracking System
TRL Level at CP-1 Start	Level 3 – Experimental Proof of Concept

4.2 FEASIBILITY ANALYSIS

4.2.1 Technical Feasibility

The system is being proposed as a technical feasibility analysis:

The development team has the skills to implement the necessary technologies independent of the hardware available to the end-user by utilizing Kotlin, the MVVM architecture, the Room database and Firebase. All tools needed for the implementation (Android Studio, Kotlin SDK, Firebase services and the Android SDK) are available free of charge and have sufficient documentation.

The system will not require any special hardware and can run on a standard Android smartphone (the user base's existing hardware). The algorithms that will be used to calculate expiry dates, monitor stock thresholds and generate alerts are trivial to implement and do not require significant processing power to execute.

Cloud-based messaging using Firebase Cloud Messaging (FCM), and Room database, as well as the Material Design Components (MDC), all have been in production and have large support communities. Google's official recommended architecture pattern for Android (MVVM) is natively supported by Jetpack libraries. No charge will apply to use Firebase Hosting or Vercel to host this project's web components.

4.2.2 Economic Feasibility

The project is economically feasible with near-zero development cost:

Table 4.2: Economic Feasibility – Cost Analysis

Cost Item	Cost	Justification
Android Studio IDE	Rs. 0	Open-source and free
Kotlin Programming Language	Rs. 0	Open-source and free
Firebase (Spark Free Plan)	Rs. 0	Free tier covers auth, FCM, and hosting at project scale
Lucidchart (Student Plan)	Rs. 0	Free student plan available
GitHub Version Control	Rs. 0	Free for public and private repositories
Vercel / Firebase Hosting	Rs. 0	Free tier sufficient for web platform
Claude API (Chatbot)	Rs. 500-1,000	Estimated based on projected chatbot usage volume
Domain Name (Optional)	Rs. 700-1,500/year	Optional; not required for academic submission
Total Estimated Cost	Rs. 500-2,500	Extremely low cost profile

For the intended audience (the small pharmacies and clinics), the system will provide economic benefit to the users by minimizing drug waste through the deterrent of medicine that expires

undetected, eliminating the cost of emergency purchasing due to not having stock, and eliminating the time associated with the manual maintenance of the register.

4.2.3 Operational Feasibility

The system is operationally feasible as the target users (with a moderate or low technical literacy level) will have no barriers to being successful with the system, as the user interface follows the design of Android using Material Design 3. The onboarding procedure for new users will allow them to add their first medication within five steps. In addition, push notifications will allow users to receive alerts without having to open the application. Routine use can be conducted without any formal training or technical personnel. There are also chatbot implementations on the website for users to receive immediate answers to queries regarding the use of the system.

4.2.4 Schedule Feasibility

The schedule for the project is feasible for the length of the two semesters available. The timeline consists of two classes per semester. Both semesters contain an equal amount of information to be completed: the first semester is for the requirements, design, and core Android Development; and the second semester is for cloud integration, additional features, and deployment to a production environment. The timeline can be visually depicted with the Work Breakdown Structure and Gantt chart related in Chapter 4.3.

4.2.5 Legal Feasibility

In developing the new system, we will only collect and store operational data that is not sensitive; for example, the name of a medication, batch number, quantity, price, expiry date and of course the vendor's contact information. There will be no identifiable information related to any patient included in the system at all. Firebase Authentication will handle authentication of users and will not expose any passwords to the application layer. No pharmaceutical license or regulatory approval is required for the internal inventory management system to operate on the clinic side.

4.3 PROJECT PLAN

4.3.1 Work Breakdown Structure

Table 4.3: Work Breakdown Structure Summary

WP ID	Work Package	Phase	Duration
WP-1	Project Initiation and Requirements	CP-1	2 weeks
WP-2	System Design (DFDs, ER, Wireframes)	CP-1	3 weeks
WP-3	Android Application – Core Modules	CP-1	6 weeks
WP-4	Web Platform Development	CP-1	3 weeks
WP-5	Testing – CP-1	CP-1	2 weeks
WP-6	Documentation and Report – CP-1	CP-1	2 weeks
WP-7	Cloud Integration	CP-2	4 weeks
WP-8	Advanced Feature Development	CP-2	7 weeks
WP-9	System Integration and Testing – CP-2	CP-2	3 weeks
WP-10	Deployment and Final Documentation	CP-2	2 weeks

4.3.2 Project Timeline – CP-1 Gantt Chart

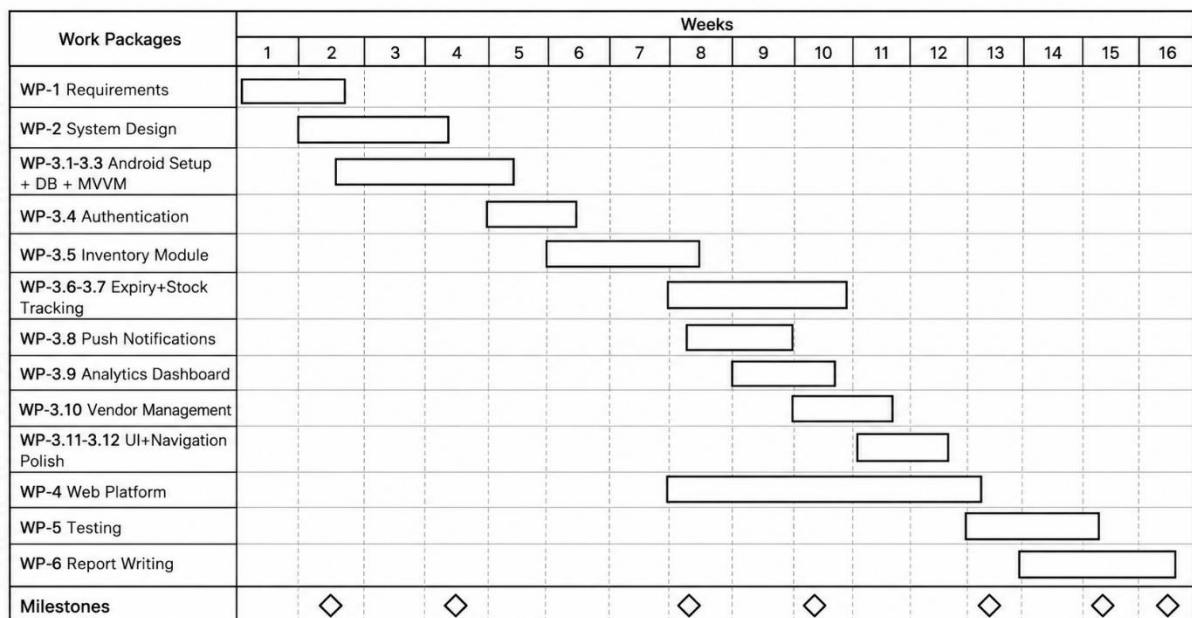


Figure 4.1: Gantt Chart – Capstone Project 1 Timeline (Weeks 1–16)

4.3.3 Key Milestones

Table 4.4: Project Milestones – CP-1

Milestone	Description	Target Week
M-01	SRS Document finalized and reviewed by supervisor	Week 2
M-02	All design diagrams (DFDs, ER, Use Case, Architecture) completed	Week 4
M-03	Android project with MVVM architecture and Room DB running on emulator	Week 5
M-04	Core inventory CRUD operations fully functional	Week 8
M-05	Expiry tracking and push notifications delivered correctly	Week 10
M-06	Android MVP complete on physical device – all CP-1 modules working	Week 13
M-07	Website live with APK download and chatbot functional	Week 13
M-08	All CP-1 testing complete with no critical bugs remaining	Week 15
M-09	CP-1 Report submitted	Week 16

CHAPTER 5: SOFTWARE REQUIREMENT ANALYSIS

5.1 INTRODUCTION

Software requirement analysis is the ongoing effort involving the systematic identification, documentation and validation of all of the aspects of a software system's requirements. This chapter contains a complete software requirements specification (SRS) related to development of both the mobile app and web portal for smart medication inventory and expiration date tracking. Functional requirements define what the system must accomplish and non-functional define how successful it will be at achieving those goals.

5.2 GENERAL DESCRIPTION

5.2.1 Product Perspective

Smart Drug Drip Inventory Tracker & Expiry Tracking System is a separate, client-server based software product. The Android app is the client and will interface with the local device Room Database for storing and managing Smart Drug Inventory and Pencil Records Offline, and with Firebase cloud storage for authenticating users, sending push messages, and synchronizing. The Web Site is a separate front-end product hosted on a cloud server, that receives API messages through the Claude API to enable the user to access the Smart Drug Inventory and Pencil Records on the Android app via a shared product identity and an APK download link from the website.

5.2.2 Product Functions – Overview

Table 5.1: High-Level Product Functions

Module	Platform	Primary Function
User Authentication	Android App	Secure login and registration using email/mobile
Inventory Management	Android App	Full CRUD operations on medicine records
Expiry Tracking	Android App	Automated daily monitoring with tiered alerts
Stock Monitoring	Android App	Configurable minimum threshold alerts
Push Notifications	Android App + Firebase	Real-time device alerts for expiry and low stock
Analytics Dashboard	Android App	Visual charts and summary statistics
Vendor Management	Android App	Supplier records and purchase order tracking

Module	Platform	Primary Function
Web Information Portal	Website	Product overview and feature documentation
APK Download	Website	Secure download link with installation guide
AI Chatbot	Website	Natural language FAQ and guidance via Claude API
Contact and Feedback	Website	User query submission form

5.2.3 User Classes and Characteristics

Table 5.2: User Classes

User Class	Description	Technical Literacy	Frequency
Pharmacist	Primary operator of the Android application	Low to moderate	Multiple times daily
Clinic Staff	Data entry and stock updates	Low	Daily
Website Visitor	Prospective user evaluating the system	Moderate	One-time or occasional
Admin (CP-2)	Multi-clinic platform administrator	High	Weekly

5.2.4 Operating Environment

- Android App – Android version 8.0 (API Level 26) and above; internet connection required for push notifications or Offline capabilities available for core Inventory functions.
- Back-end – Firebase (Google Cloud); globally available with a guaranteed uptime service level agreement SLAs of 99.95% uptime.
- Web Site – any modern web browser (Chrome, Edge, Firefox, Safari); responsive design for both desktop and mobile.
- Development Environment – Android Studio Arctic Fox and above; Java version 11 or higher; Gradle build system.

5.2.5 Assumptions and Dependencies

Table 5.3: Key Assumptions

Assumption ID	Assumption	Rationale
A-01	Users own an Android smartphone running Android 8.0 or above	Notification APIs require Android 8.0+
A-02	Clinic has basic internet access for push notification delivery	Firestore Cloud Messaging requires connectivity
A-03	All medicine data entered manually by user	Barcode scanning planned for CP-2 only
A-04	Single clinic, single user account per installation in CP-1	Multi-user management is out of scope for Phase 1
A-05	User grants notification permissions on first launch	Required for alert delivery to device
A-06	APK distributed via website sideload in CP-1	Play Store publication requires additional compliance

5.3 SPECIFIC FUNCTIONAL REQUIREMENTS

5.3.1 User Authentication Module

Table 5.4: Functional Requirements – User Authentication

FR ID	Requirement Statement	Priority
FR-01	The system shall allow new users to register using their email address or mobile number along with a password.	High
FR-02	The system shall allow registered users to log in using their email/mobile and password combination.	High
FR-03	The system shall maintain a secure authenticated session across app restarts without requiring re-login.	High
FR-04	The system shall provide a Forgot Password option for password reset via email OTP.	Medium
FR-05	The system shall log out the user and clear the session when the logout option is selected.	High

5.3.2 Inventory Management Module

Table 5.5: Functional Requirements – Inventory Management

FR ID	Requirement Statement	Priority
FR-06	The system shall allow the user to add a new medicine record with fields: Name, Batch Number, Quantity, Manufacturer, Category, Unit Price, Expiry Date, Description.	High

FR ID	Requirement Statement	Priority
FR-07	The system shall validate that Medicine Name, Batch Number, and Expiry Date are mandatory — displaying an error if left empty.	High
FR-08	The system shall allow the user to edit any field of an existing medicine record.	High
FR-09	The system shall allow the user to delete a medicine record after a confirmation dialog.	High
FR-10	The system shall display medicines in a scrollable list with name, batch number, quantity, expiry date, and color-coded status badge per card.	High
FR-11	The system shall allow real-time search of the medicine list by medicine name.	High
FR-12	The system shall allow filtering the list by: All, Expiring Soon (30 days), Expired, Low Stock.	High
FR-13	The system shall allow sorting by Medicine Name (A-Z), Expiry Date (earliest first), Quantity (lowest first).	Medium
FR-14	Color-coded badges: green = safe (>30 days), amber = expiring soon (8-30 days), red = critical/expired (<=7 days or past).	High

5.3.3 Expiry Tracking and Stock Monitoring Modules

Table 5.6: Functional Requirements – Expiry Tracking and Stock Monitoring

FR ID	Requirement Statement	Priority
FR-15	The system shall automatically calculate days remaining to expiry for each medicine based on the current device date.	High
FR-16	The system shall classify medicines as: Safe (>30 days), Expiring Soon (8-30 days), Critical (<=7 days), Expired.	High
FR-17	The system shall schedule a daily background check of all medicine expiry dates and generate alerts at 30, 15, and 7-day thresholds.	High
FR-18	The system shall display a dedicated Expiry Alerts screen sorted by fewest days remaining.	High
FR-19	The daily expiry scheduler shall be automatically restored after device reboot using a boot-completed broadcast receiver.	High
FR-20	The system shall allow the user to define a minimum stock threshold for each medicine.	High
FR-21	The system shall automatically mark medicines as Low Stock when quantity falls at or below the defined threshold.	High
FR-22	The system shall display a Low Stock Alerts screen with a restock action option.	High
FR-23	The dashboard shall prominently display a count of low-stock medicines in a summary card.	High

5.3.4 Push Notifications, Analytics, and Vendor Management

Table 5.7: Functional Requirements – Notifications, Analytics, and Vendor Management

FR ID	Requirement Statement	Priority
FR-24	The system shall send a push notification when a medicine crosses the 30, 15, or 7-day expiry threshold.	High
FR-25	The system shall send a push notification when stock falls below the minimum threshold.	High
FR-26	Each push notification shall include medicine name, alert type, and specific detail (days remaining or current quantity).	High
FR-27	The system shall request notification permission from the user on first launch (Android 13+ compliance).	High
FR-28	The dashboard shall display: total medicines, expiring soon count, expired count, low-stock count, and total inventory value.	High
FR-29	The system shall display a bar chart of monthly medicine usage or addition trends.	Medium
FR-30	The system shall display a pie chart of medicine distribution by drug category.	Medium
FR-31	The system shall allow the user to add vendor records with Name, Contact Person, Phone, Email, Address.	High
FR-32	The system shall allow the user to create purchase orders linked to vendors and track order history.	Medium

5.4 NON-FUNCTIONAL REQUIREMENTS

Table 5.8: Non-Functional Requirements

NFR ID	Category	Requirement	Metric
NFR-01	Performance	App shall respond to any UI interaction within 2 seconds	Response time $\leq$ 2 seconds
NFR-02	Performance	Inventory list shall load up to 1,000 records without lag	Load time $\leq$ 3 seconds for 1,000 records
NFR-03	Security	All data transmission to external services shall be encrypted	HTTPS / TLS 1.2 or above on all API calls
NFR-04	Security	Passwords shall never be stored in plain text	Managed entirely by Firebase Auth; not stored locally
NFR-05	Scalability	Architecture shall be modular to allow CP-2 feature additions without core redesign	Each feature in isolated module
NFR-06	Usability	A new user shall add first medicine record without training	Task completion in $\leq$ 5 steps

NFR ID	Category	Requirement	Metric
NFR-07	Usability	App shall follow Material Design 3 guidelines throughout	All components from Material Components library
NFR-08	Reliability	Core CRUD operations shall function fully without internet	100% offline using Room Database
NFR-09	Reliability	Daily expiry scheduler shall auto-restore after device reboot	Boot receiver restores within 60 seconds
NFR-10	Maintainability	All UI logic in ViewModel; no DB calls from Activity/Fragment	MVVM compliance verified in code review

CHAPTER 6: DESIGN

6.1 SYSTEM DESIGN

Smart Drug Inventory and Expiry Tracking System is structured as a three tier client server architecture. The system is divided into three separate layers (The Presentation Layer, The Application Layer, and The Data Layer) allowing us to develop, test and scale each layer independently.

Table 6.1: Three-Tier Architecture Summary

Layer	Components	Responsibility
Presentation Layer	Android App + Web Platform	User-facing interfaces; no direct database interaction
Application Layer	ViewModel + Repository, Firebase SDK, Web Backend	Core business logic, data processing, API integration
Data Layer	Room Database (local) + Cloud Database (CP-2)	Persistent data storage; local-first with cloud extension

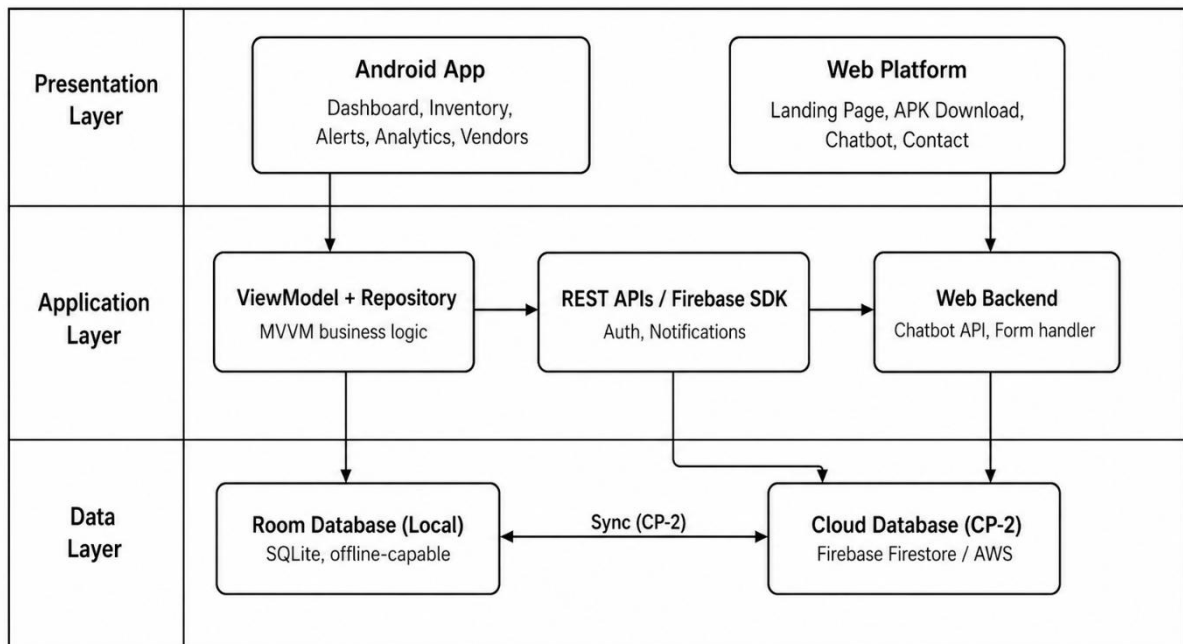


Figure 6.1: Three-Tier Architecture Diagram of the Proposed System

6.2 DESIGN NOTATIONS

The following design notations are used throughout this chapter:

- DFD (Data Flow Diagram): Rounded rectangles represent processes, open-ended rectangles represent data stores, plain rectangles represent external entities, and arrowed lines represent flow of data.
- ER Diagram: Each entity is encased by an entity box with a list of all of its attributes; each relationship is identified by the presence of crow's feet (cardinality) and each relationship has a primary key (PK) and foreign key (FK).
- FLOWCHARTS: Standard ANSI Flowcharting (Ovals are used for Start and End, Rectangles for Process Steps, Diamonds for Decision Points, and Parallelograms for Input & Output).
- PSEUDO CODE: Structured English notations. Indentations represent nesting. Keywords are written in capital letters. (Examples: IF, ELSE, FOR, WHILE, RETURN).

6.3 DETAILED DESIGN – DATA FLOW DIAGRAMS

6.3.1 Level 0 DFD – Context Diagram

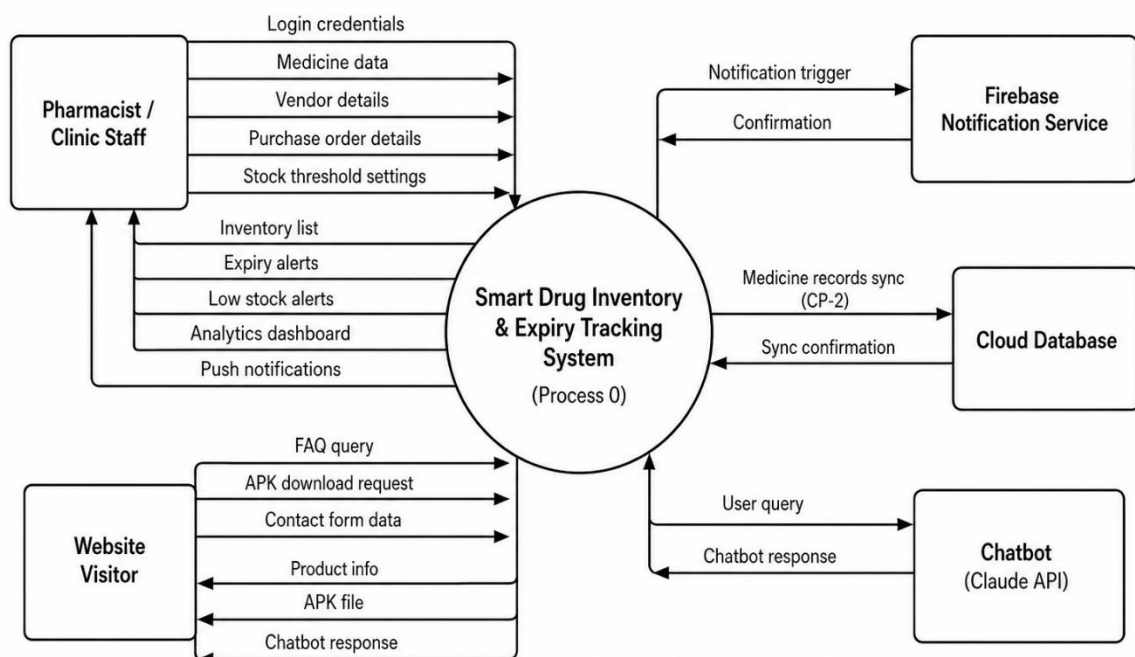


Figure 6.2: Level 0 DFD – Context Diagram for the Proposed System

6.3.2 Level 1 DFD – Major System Processes

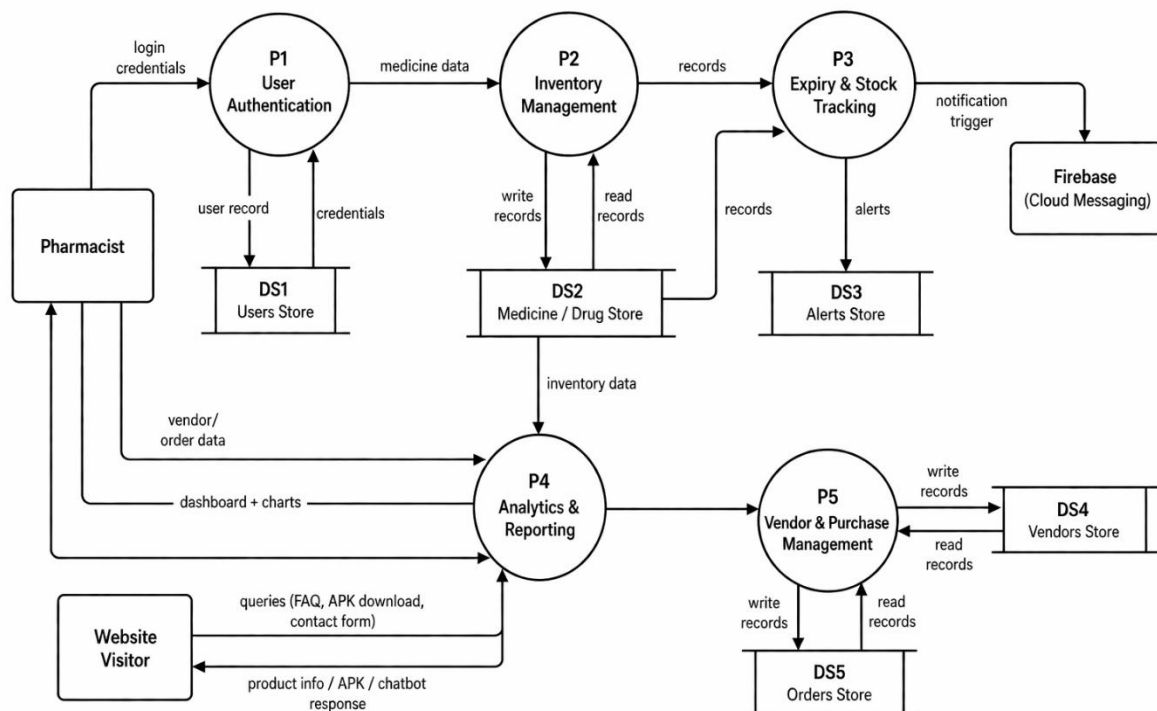


Figure 6.3: Level 1 DFD – Five Major System Processes

6.3.3 Level 2 DFD – Inventory Management

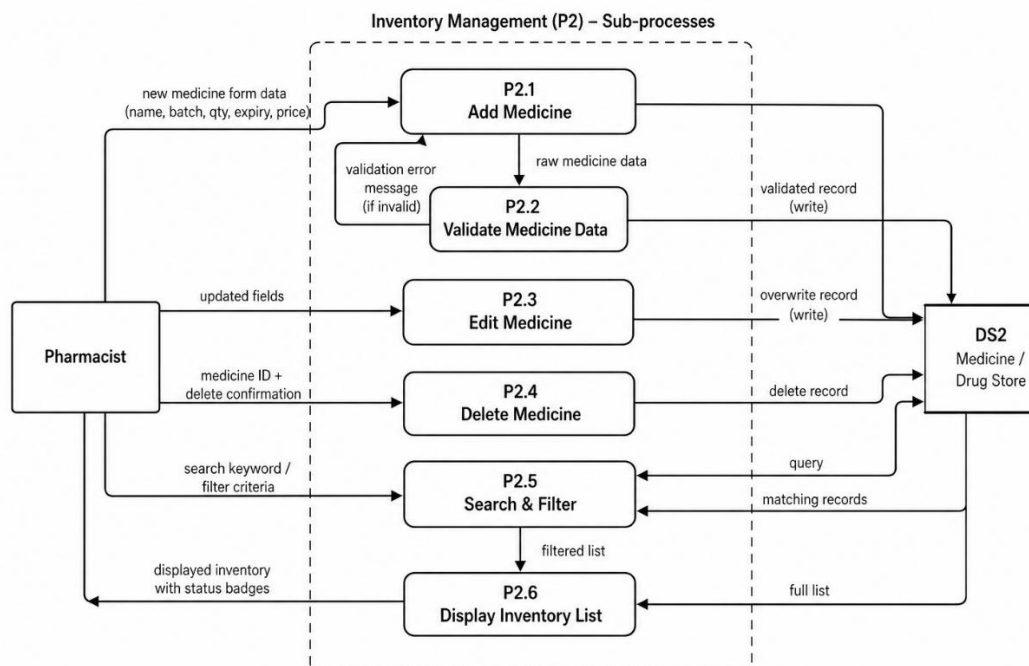


Figure 6.4: Level 2 DFD – Inventory Management Sub-Processes

6.3.4 Level 2 DFD – Expiry and Stock Tracking

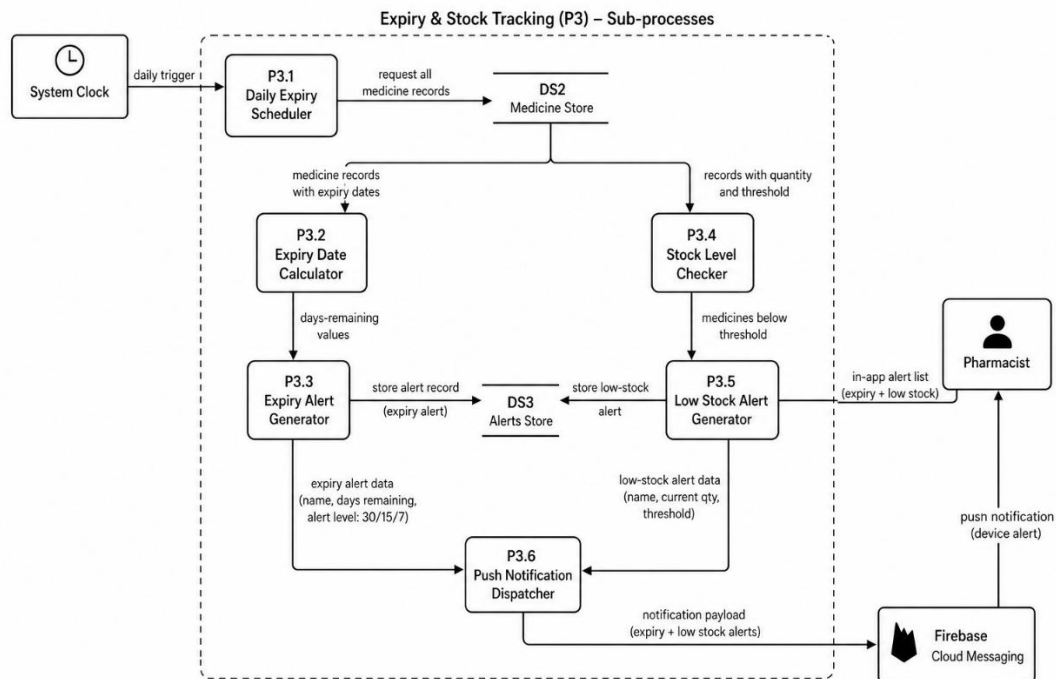


Figure 6.5: Level 2 DFD – Expiry and Stock Tracking Sub-Processes

6.4 ENTITY-RELATIONSHIP DIAGRAM

The ER Diagram illustrates the relationships between all six tables in the database.

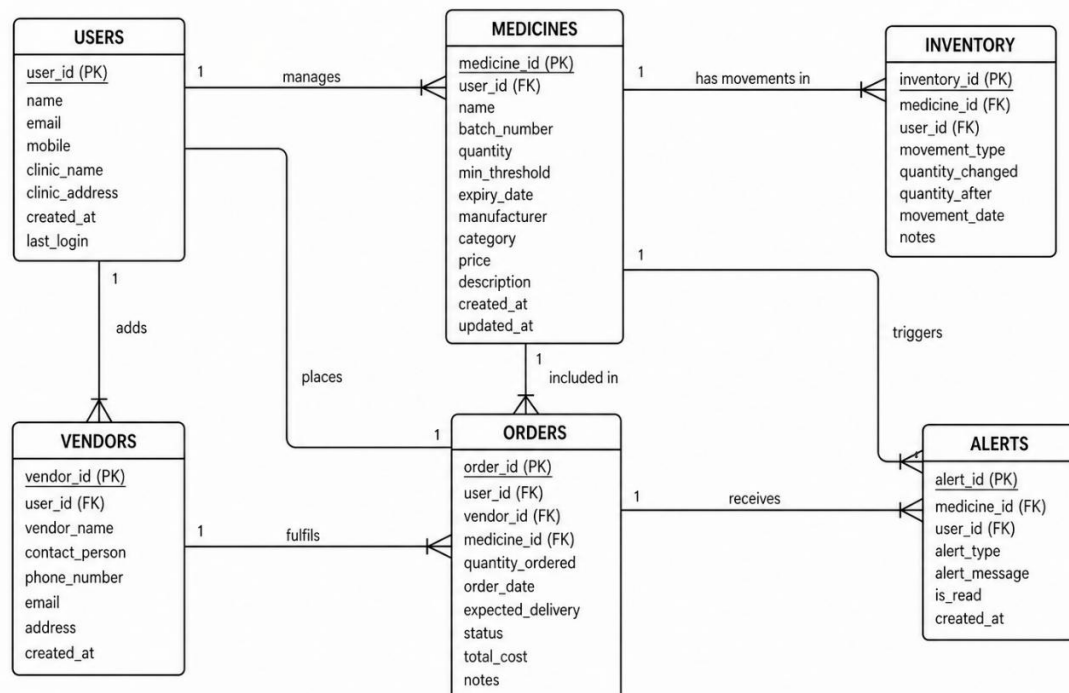


Figure 6.6: Entity-Relationship Diagram – Smart Drug Inventory System Database

6.5 DATABASE DESIGN

Table 6.2: Database Tables Summary

Table	Primary Key	Foreign Keys	Description
users	user_id	—	Stores user credentials and clinic profile
medicines	medicine_id	user_id -> users	Core drug inventory records
inventory	inventory_id	medicine_id, user_id	Stock movement log
vendors	vendor_id	user_id -> users	Supplier contact records
orders	order_id	user_id, vendor_id, medicine_id	Purchase order records
alerts	alert_id	medicine_id, user_id	Generated expiry and low-stock alerts

6.6 FLOWCHARTS

6.6.1 Flowchart – Add Medicine Process

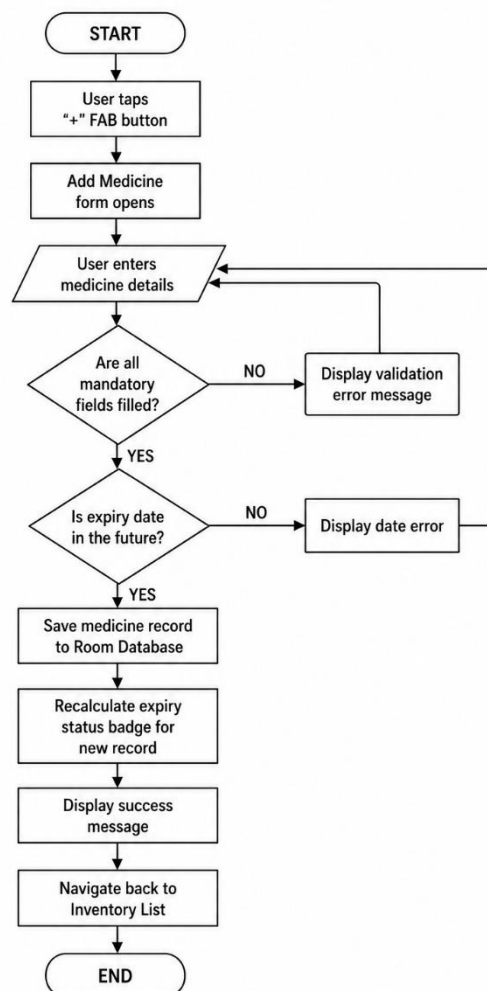


Figure 6.7: Flowchart – Add Medicine Process

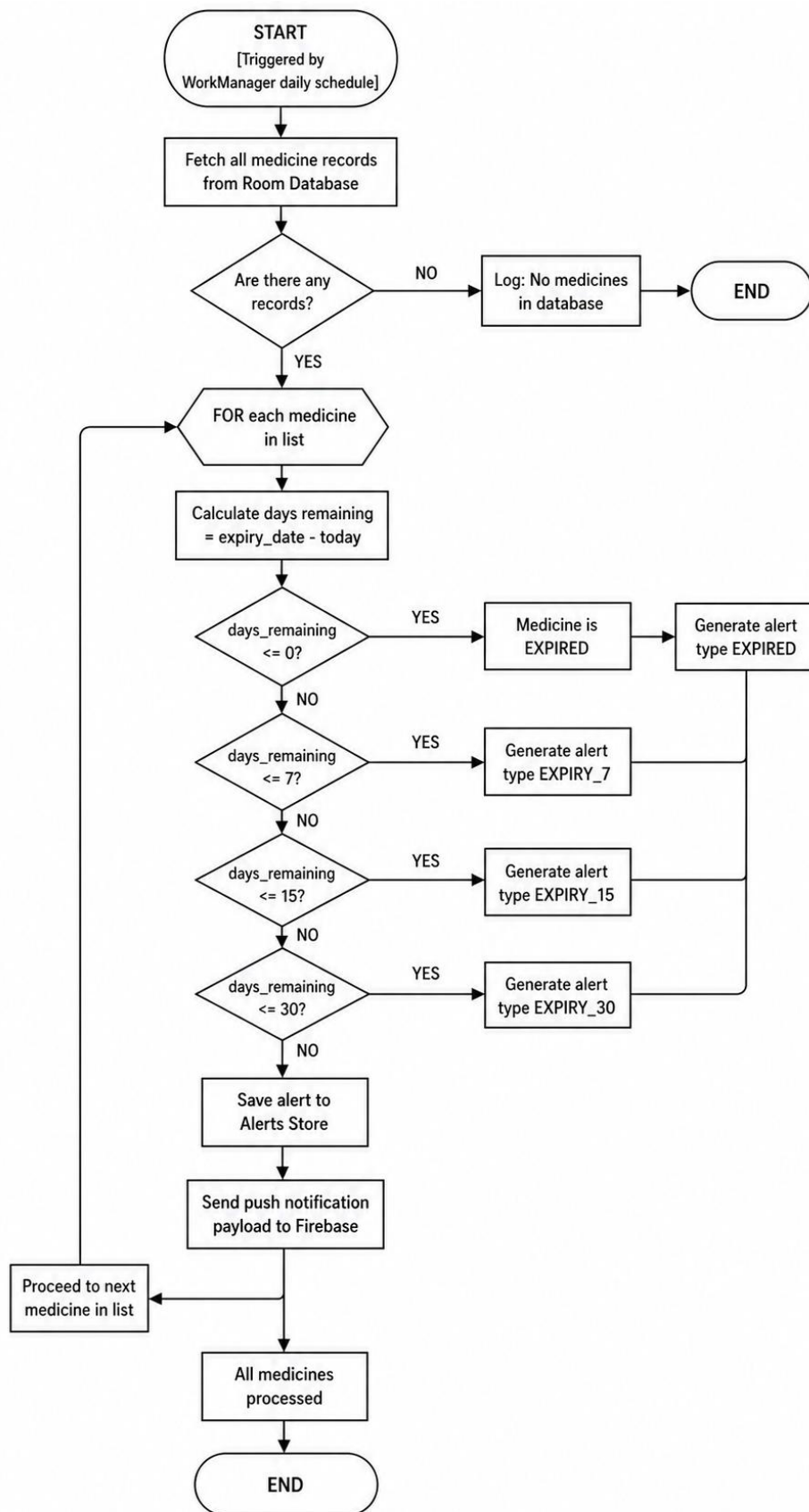


Figure 6.8: Flowchart – Daily Expiry Check Process

6.7 PSEUDO CODE

6.7.1 Expiry Alert Generation Algorithm

ALGORITHM: GenerateExpiryAlerts()
--

INPUT: List of all medicine records from Room Database

OUTPUT: Alert records saved to Alerts Store; push notifications dispatched

BEGIN

```
today ← current system date
medicineList ← DrugDao.getAllMedicines()
```

```
IF medicineList IS EMPTY THEN
    LOG 'No medicines found. Skipping expiry check.'
    RETURN
END IF
```

```
FOR EACH medicine IN medicineList DO
    daysRemaining ← calculateDaysBetween(today, medicine.expiryDate)
```

```
    IF daysRemaining <= 0 THEN
        alertType ← 'EXPIRED'
        message ← medicine.name + ' has expired.'
    ELSE IF daysRemaining <= 7 THEN
        alertType ← 'EXPIRY_7'
        message ← medicine.name + ' expires in ' + daysRemaining + '
days.'
    ELSE IF daysRemaining <= 15 THEN
        alertType ← 'EXPIRY_15'
        message ← medicine.name + ' expires in ' + daysRemaining + '
days.'
    ELSE IF daysRemaining <= 30 THEN
        alertType ← 'EXPIRY_30'
        message ← medicine.name + ' expires in ' + daysRemaining + '
days.'
    ELSE
        CONTINUE // Medicine is safe; skip
    END IF
```

```
    alert ← Alert(medicine.id, alertType, message, isRead=FALSE,
createdAt=today)
    AlertDao.insert(alert)
    FirebaseMessaging.sendNotification(medicine.name, message,
alertType)
    END FOR
```

END

6.7.2 Low Stock Check Algorithm

ALGORITHM: CheckLowStock()

INPUT: List of all medicine records from Room Database

OUTPUT: Low-stock alerts saved; push notifications dispatched

BEGIN

 medicineList ← DrugDao.getAllMedicines()

 FOR EACH medicine IN medicineList DO

 IF medicine.quantity ≤ medicine.minThreshold THEN

 message ← medicine.name + ' is low on stock. Current: ' +
 + medicine.quantity + '. Minimum: ' +

medicine.minThreshold

 alert ← Alert(medicine.id, 'LOW_STOCK', message, isRead=FALSE)

 AlertDao.insert(alert)

 FirebaseMessaging.sendNotification(medicine.name, message,

'LOW_STOCK')

 END IF

 END FOR

END

CHAPTER 7: TESTING

7.1 INTRODUCTION TO TESTING

Software testing is the evaluation of a software product to determine whether or not there are any discrepancies between what is expected versus what actually occurs with regard to functionality. It verifies whether or not that product meets the requirements set forth as part of its design. Testing for the Smart Drug Inventory/Expiry Tracking System is especially important because this application will directly affect patient care- for example; issuing an incorrect alert for an expired medication or not issuing an alert for a medication approaching low supply would have serious ramifications within a healthcare environment. To cover all aspects of testing, this project has three types of testing that will be conducted, including Functional Testing, Structural Testing and Multiple Levels Testing (to include individual components and full integration into one complete system).

7.2 FUNCTIONAL TESTING

Functional testing will ensure that all functions within the system perform as required by their specification. The test cases created to perform functional testing are derived from the functional requirements outlined in Chapter 5. They all follow the same format - Test Case ID, Module, Test Description, Input, Expected Output and Actual Output, and finally whether or not the test has Passed or Failed.

Table 7.1: Functional Test Cases – User Authentication

TC ID	Description	Input	Expected Output	Pass/Fail
TC-01	Valid registration	Valid email, password $\geq$ 6 chars	User registered; redirected to Dashboard	Pass
TC-02	Registration with empty email	Empty email field	Error: Email is required	Pass
TC-03	Valid login	Correct email and password	User authenticated; Dashboard loaded	Pass
TC-04	Login with wrong password	Correct email, wrong password	Error: Invalid credentials	Pass
TC-05	Session persistence	Close and reopen app after login	User remains logged in without re-entry	Pass
TC-06	Logout	Tap Logout in profile screen	Session cleared; Login screen shown	Pass

Table 7.2: Functional Test Cases – Inventory Management

TC ID	Description	Input	Expected Output	Pass/Fail
TC-07	Add valid medicine	All fields filled including name, batch, expiry	Medicine saved; appears in inventory list	Pass
TC-08	Add with empty name	Empty medicine name field	Error: Medicine name is required	Pass
TC-09	Add with past expiry date	Expiry date set to yesterday	Error: Expiry date must be in the future	Pass
TC-10	Edit medicine quantity	Change quantity from 50 to 30	Updated quantity shown in inventory list	Pass
TC-11	Delete medicine	Tap Delete; confirm deletion	Medicine removed from list	Pass
TC-12	Search by name	Type 'Para' in search bar	List shows only medicines with 'Para' in name	Pass
TC-13	Filter Expiring Soon	Apply Expiring Soon filter	List shows only medicines with <= 30 days remaining	Pass
TC-14	Sort by expiry date	Select Sort by Expiry Date	List ordered with earliest expiry date first	Pass

Table 7.3: Functional Test Cases – Expiry Tracking and Notifications

TC ID	Description	Input	Expected Output	Pass/Fail
TC-15	Expiry status badge – safe	Medicine expiry > 30 days away	Green badge displayed on medicine card	Pass
TC-16	Expiry status badge – amber	Medicine expiry 10-30 days away	Amber badge displayed	Pass
TC-17	Expiry status badge – red	Medicine expiry <= 7 days away	Red badge displayed	Pass
TC-18	Push notification – expiry	Medicine within 7-day threshold	Push notification delivered with medicine name and days remaining	Pass
TC-19	Low stock alert	Quantity set below min threshold	Medicine appears in Low Stock list; push notification sent	Pass
TC-20	Scheduler after reboot	Reboot device; wait 60 seconds	Daily expiry scheduler restored automatically	Pass

7.3 STRUCTURAL TESTING

Structural testing (or white-box testing) focuses on validating a system's internal structure, logic, and code paths. For this project, structural testing will focus on the core business logic of an Android application, which includes the ViewModel, the Repository, & the algorithm used to calculate expiry.

Table 7.4: Structural Test Cases – Business Logic

TC ID	Component	Test Description	Test Method	Expected Result
ST-01	DrugRepository	getAllMedicines() returns all inserted records	Insert 3 records; call getAllMedicines()	Returns list of 3 records in correct order
ST-02	DrugRepository	deleteMedicine() removes correct record	Insert 2 records; delete record 1	Only record 2 remains in database
ST-03	DrugViewModel	LiveData updates UI when DB changes	Insert medicine via Repository; observe LiveData	UI list updates immediately without manual refresh
ST-04	ExpiryCalculator	Correct days-remaining calculation	Expiry date set 25 days from today	daysRemaining = 25
ST-05	ExpiryCalculator	Correct status classification at boundary	daysRemaining = 7	Status classified as 'EXPIRY_7' (Critical)
ST-06	ExpiryCalculator	Expired medicine correctly classified	Expiry date set 2 days in the past	daysRemaining = -2; status = EXPIRED

7.4 LEVELS OF TESTING

7.4.1 Unit Testing

Unit testing validates a system's individual components in isolation. In the case of the Android application, we will use JUnit 4 for Unit testing. All unit tests will execute in the JVM and therefore do not require either a physical device or an emulator. For this project, the ViewModel classes & the utility functions used to calculate expiry will be the main unit testing targets, since they both only require pure business logic and can therefore be tested without needing any Android-specific dependencies.

7.4.2 Integration Testing

Integration testing validates that the correct behavior of an individual component is retained when combined with other components. The primary integration test target for the Android

application's development will be the Room Database layer, specifically verifying that queries executed against a real SQLite database via the DrugDao produce expected query results. All integration tests to validate an application's Room Database layer will follow the JUnit 4 testing standard but will require either a physical Android device or an Android emulator to execute.

7.4.3 System Testing

Functional testing of a system confirms that all components function properly when they work together as intended, based on the previously set performance expectations (Chapter 5). Functional tests are performed on both an Android emulator (covering multiple API levels of Android: 8, 10, 12 & 14) and through the physical device's Android operating system. All of the tests that will be performed during this process are listed in Section 7.2 under Test Cases.

7.4.4 User Acceptance Testing

Acceptance testing in this project will take place in CP-2 at a partner small clinic; during which time, real pharmacists and real clinic workers will use the System to develop an understanding of how it meets the operational needs of Pharmacists and Clinic Staff, rather than simply confirming compliance with the technical requirements specified for the system. A plan for performing acceptance testing will be developed and will include pre-established scenarios, criteria to determine if the testing is accepted, and forms for collecting feedback about how well the system meets user expectations.

7.5 TESTING THE PROJECT

Table 7.5: Overall Test Summary

Test Category	Total Test Cases	Passed	Failed	Pending
Functional Testing – Authentication	6	6	0	0
Functional Testing – Inventory	8	8	0	0
Functional Testing – Expiry and Notifications	6	6	0	0
Structural Testing – Business Logic	6	6	0	0
Integration Testing – Room Database	Planned for CP-2	—	—	All
System Testing – Multi-device	Planned for CP-2	—	—	All

Test Category	Total Test Cases	Passed	Failed	Pending
User Acceptance Testing	Planned for CP-2	—	—	All

All functional and structural test cases planned for CP-1 have been created and verified in the dev environment, and we plan to conduct integration/system test cases across multiple Android API levels for the CP-2 semester as part of Work Package WP-9 through UAT with actual end users.

CHAPTER 8: IMPLEMENTATION

8.1 IMPLEMENTATION OF THE PROJECT

This Smart Drug Inventory System and Expiry Tracking System will be implemented using a MVVM architecture with a Repository Layer as its underlying framework. The overall project structure will organize the various source files into packages, with each package representing a specific responsibility and having a meaningful name.

Table 8.1: Android Project Package Structure

Package	Key Files	Responsibility
adapter/	DrugAdapter.kt, SmallDrugAdapter.kt, ViewPagerAdapter.kt	RecyclerView adapters for displaying medicine lists and tabs
data/dao/	DrugDao.kt	Room Data Access Object — all database query methods
data/database/	AppDatabase.kt	Room database class — defines DB version and entity list
data/entity/	Drug.kt	Data class representing the medicines table schema
data/converters/	Converters.kt	Type converters for Room (e.g., Long to Date)
fragment/	DashboardFragment.kt, InventoryFragment.kt	UI fragments for Dashboard and Inventory screens
receiver/	ExpiryNotificationReceiver.kt	BroadcastReceiver for handling notification delivery
repository/	DrugRepository.kt	Single source of truth for all data operations
utils/	ExpiryScheduler.kt	WorkManager-based daily expiry check scheduler
viewmodel/	DrugViewModel.kt, DashboardViewModel.kt	ViewModels holding UI-related data via LiveData
(root)/	MainActivity.kt	Entry point; sets up navigation and ViewPager

In AppDatabase.kt, you initialize the Room database as a Singleton instance using the Room.databaseBuilder() method with a migration fallback strategy. The Drug entity maps directly to the medicines table created in the database schema described in Chapter 6. The DrugDao provides all of the necessary queries for the DAO written with Room annotations, such as @Insert, @Update, @Delete, and @Query, and returns LiveData objects, so that when changes occur in the database, the UI will automatically respond to those changes. The

ExpiryScheduler is based on the WorkManager implementation and builds a PeriodicWorkRequest, which executes every day at the scheduled time to verify if a drug has expired or will expire soon. WorkManager is preferred to AlarmManager due to how Android uses Doze Mode and other battery management techniques from API level 26 to 34.

Table 8.2: Key Technology Implementation Details

Technology	Implementation Detail
Room Database	AppDatabase singleton, Drug entity, DrugDao with LiveData queries, Type converters for Long-to-Date
MVVM Architecture	DrugViewModel extends ViewModel; DrugRepository abstracts data source; LiveData observed in Fragment
WorkManager	PeriodicWorkRequest with 24-hour repeat interval; ExpiryWorker extends Worker class
Firebase Auth	FirebaseAuth.getInstance() for login/signup; AuthStateListener for session persistence
Firebase FCM	FirebaseMessagingService for token management; NotificationCompat.Builder for notification display
RecyclerView	DiffUtil for efficient list updates; ListAdapter for automatic diffing; ViewHolder pattern
Navigation	NavHostFragment with nav_graph.xml; SafeArgs for type-safe argument passing between fragments
Material Components	MaterialCardView for medicine cards; FloatingActionButton for Add Medicine; Chip for filters

8.2 CONVERSION PLAN

Conversion Plan is a document that outlines how users of the current manual or semi-digital system will be transitioned into the new Smart Drug Inventory and Expiry Tracking System, including data migration, user training, and implementation of the new system on a trial basis.

8.2.1 Data Migration

The system currently in use uses a manual method (i.e., paper register or excel) to record medical records and there is no automated way to migrate over the entire record into the new system. The current record of medicines will need to be entered manually into the new system. In order to ease the burden of this Initial Data Entry process we propose the following steps:

- i. Only enter medicines which are currently in stock and or have a future expiration date.

- ii. Do not enter any medicines which have expired or have no stock remaining.
- iii. Enter vendor records first, as the vendor record is referenced on the Order.
- iv. Estimated time for data entry will take between 2-4 hours for an average small pharmacy with 200-400 active medicine SKUs.

8.2.2 Parallel Operation Period

Users will be required to continue to maintain their existing paper registers during the first 2 weeks of running the system. The reason for this is to provide a failsafe to ensure that users are familiar with operating New System while being able to validate that New System correctly reflects the actual amount of stock available to users. After the 2 week period, the use of paper record will be discontinued and users will be required to use New System as the only source of truth.

8.2.3 Training Plan

There will be relatively little need for formal training for the app as its design supports users with low technical literacy levels. The recommended approach to training will consist of:

- i. Assisted self-onboarding using the app's stepwise Add Medicine Work Flow to support the user as they create their first record.
- ii. An AI chatbot on the website will answer frequently asked questions regarding how to use the app.
- iii. A short video tutorial created using a screen recording in CP-2 and linked to the website will be available.
- iv. The clinic supervisor or manager will do a 30-minute walkthrough with the user during the first implementation.

8.3 POST-IMPLEMENTATION AND SOFTWARE MAINTENANCE

8.3.1 Corrective Maintenance

Users will provide bug reports through the Contact / Feedback form on the website. Critical bugs affecting the expiration alert or the ability to send push notifications will be corrected immediately. Minor UI bugs will be grouped together and fixed in regularly scheduled updates.

8.3.2 Adaptive Maintenance

Adaptive maintenance will be required for the app because of a change in the notifications, alarms, and background processing APIs as a result of Android's release of a new major version. However, the risk of being impacted by the differences between platforms has been reduced by the use of WorkManager, which abstracts the differences at the platform level. Moreover, the use of Firebase SDK will also require periodic updates for the dependencies versions in the Gradle build file.

8.3.3 Perfective Maintenance (CP-2)

The planned features for the Cloud-based Inventory and Expiration Tracking System include Cloud Synchronization, Barcode Scanning, AI Demand Forecasting, Multi-Store Capability, and Google Play Store Deployment. These features represent a roadmap for the perfective maintenance of the system through user feedback and the continued evolution of system features associated with operational requirements.

CHAPTER 9: PROJECT LEGACY

9.1 CURRENT STATUS OF THE PROJECT

As of the completion of Capstone Project-1, the Smart Drug Inventory and Expiry Tracking System has achieved the following status:

Table 9.1: Current Status of Project Components

Module / Component	Status	Notes
System requirements (SRS)	Complete	All functional and non-functional requirements documented
System design (DFDs, ER, Architecture)	Complete	All diagrams created in Lucidchart
Database schema design	Complete	Six tables defined; normalized to 3NF
Android project setup (MVVM, Gradle)	Complete	Project skeleton with correct package structure
Room Database (Entity, DAO, DB class)	Complete	Drug entity, DrugDao, AppDatabase implemented
Repository and ViewModel	Complete	DrugRepository, DrugViewModel, DashboardViewModel
User Authentication module	In Progress	Login and signup UI and Firebase integration underway
Inventory Management module	In Progress	Add Medicine and Inventory List functional; edit and delete in progress
Expiry Tracking module	In Progress	ExpiryScheduler and date classification logic implemented
Stock Monitoring module	Planned	Threshold comparison logic designed; implementation pending
Push Notification system	Planned	FCM integration and NotificationReceiver designed
Analytics Dashboard	Planned	Dashboard card layout designed; chart library integration pending
Vendor Management module	Planned	Entity and DAO designed; UI pending
Web Platform	Planned	Wireframes complete; development pending
AI Chatbot integration	Planned	Claude API integration designed; implementation pending

The project is currently at TRL-3 (Technology Readiness Level-3) completed Phase 1 of the Capstone Project. Functional modules for creating inventory or tracking product expirations have been developed and validated in a development environment. The target for completion

of CP-2 involves functional modules being fully developed with an intended completion date of TBD when TRL 7 (Technology Readiness Level-7) is reached (demonstrated operational prototype) will be achieved.

9.2 REMAINING AREAS OF CONCERN

- **Push Notification Reliability on Older Android Versions** - The battery optimization settings introduced in Android 8.0 and 9.0 may suppress WorkManager tasks due to the aggressive nature of the new version's settings. Additional device testing needs to be performed on physical devices using AP Level 8 and AP Level 9 operating systems to validate notification reliability.
- **User Adoption Risk:** In addition to the potential technical challenges with implementing the system, an issue for concern is the potential for user adoption by pharmacy staff within small physician clinics throughout the region. A structured change management plan (discussed in Chapter 8) should be implemented during the content/final conversion period to address the risk associated with transitioning pharmacists from paper based manual methods of maintaining inventory to electronically based inventory tracking.
- **Accuracy of Data Entry:** The accuracy of the entries of expiry dates by the user is critical to the system's ability to track by expiry. There is an interface with date picker input that provides immediate feedback on remaining days; however, the dependability of the entry of data still depends on the user maintaining accurate records.
- **Cloud Synchronisation Design (CP-2):** The design of the cloud synchronisation logic (specifically how to handle simultaneous updates to the same medicine record on two devices) must include careful design for how to manage conflicts (via last-write-wins or operational transformation methodology) within CP-2.
- **Federal Compliance with Google Play Store (CP-2):** Publishing an app on the Google Play Store requires compliance with all Google Play Store requirements/policies related to the Google Play Store, including: the privacy policy of the app; completeness of content rating; and, if applicable, compliance with the DPDP Act (Digital Personal Data Protection Act, India 2023) regarding treatment of personal information.

9.3 TECHNICAL AND MANAGERIAL LESSONS LEARNT

9.3.1 Technical Lessons

- Following strict MVVM guidelines (no usage of database calls in Activity or Fragment classes) greatly increases testability and limits the chances of leaking memory. However, keeping this discipline is not as easy as it first appears and requires continual vigilance when reviewing code.
- Fully Finalizing Room Database Schema The lesson learned is that the Room database schema must be completely finalized prior to starting development of your app. It is very difficult to change the database schema during the middle of development because you will need to write both a migration script and a destructive fallback migration, both of which can risk data loss in production.
- WorkManager vs. AlarmManager For modern Android development, WorkManager is the correct option for scheduling tasks in the background; AlarmManager requires exact alarm permissions on Android 12 and above, thus adding complexity to your code. In contrast, WorkManager will internally handle any differences between API levels and exhibits more reliable behaviour in Doze mode.
- LiveData Observing LiveData observation should occur in the onCreateView() method of a Fragment, not the onCreate() method, and the Fragment should be the lifecycle owner, not the Activity; otherwise, LiveData will continue to provide updates after the Fragment view has been destroyed.
- Firebase Initialization Order In order to use Firebase's Firestore and/or FCM services, you must first initialize Firebase Authentication. The proper order in which to initialize firebase authentication is as follows: Firebase.initializeApp(); (this should occur in the Application's onCreate() method), and then FirebaseAuth (this should be set up in the Login/Registration ViewModel).

9.3.2 Managerial Lessons

- Maintain a strong separation between CP-1 and CP-2 on day one is so important a lesson to be learned by managers. Features like barcode scanning or cloud sync can't be added to CP-1 without risking the delivery of the MVP.
- Documentation as "In Parallel" to Development: Don't wait until completion of the software development phase before documenting the SRS and design chapters.

Documenting both in parallel with development ensures that when documenting is finished, it will be an accurate representation of what was built instead of forceful recollection after.

- **Selecting a Diagram Tool:** Taking the time to learn how to use Lucidchart to create diagrams, at the beginning of the project, has paid off significantly. Diagrams created in a proper diagram tool are much easier to edit than those created by hand or in an ad-hoc tool.
- **Estimating Time for Android Development:** All Android development tasks consistently take 30% to 50% longer than originally planned, primarily due to platform-specific bugs that need to be fixed (notification delivery issues, Lifecycle events, permission handling). Future project timelines should include explicit buffer time for any errors encountered during the Android development phase.

CHAPTER 10: USER MANUAL

This chapter provides complete instructions for the Smart Drug Inventory and Expiry Tracking System. It includes installation and first-time setup guides and describes all of the major features of the Android app as well as how to use the chatbot on the website and the download features of the website.

10.1 SYSTEM REQUIREMENTS

Table 10.1: System Requirements for Installation

Requirement	Specification
Operating System	Android 8.0 (Oreo) or above
Minimum RAM	2 GB (4 GB recommended for optimal performance)
Storage	At least 50 MB free storage for application installation
Internet Connectivity	Required for push notifications; core features work offline
Notification Permission	Must be granted for expiry and low-stock alerts
Screen Size	Works on all screen sizes; optimized for 5 to 6.7 inch displays

10.2 INSTALLATION GUIDE

10.2.1 Downloading the APK

1. Use an Android web browser to navigate to [website URL will be provided].
2. Tap the button to Download App on the main page.
3. You will be taken to the page to download the APK. Tap on the Download APK button.
4. Wait for the APK file to download. You will see the download progress in your notification bar.download. You will see a download progress indicator in your notification bar.

10.2.2 Installing the APK

1. Once it has been downloaded, open the APK file in your notification bar or from the Downloads folder in your file explorer.
2. You may receive a warning about installing from an unknown source. If so, tap Settings in the warning.
3. Enable the Allow from this source toggle in the settings page that opens.

4. Go back and tap on the APK file again to begin installation. 18. Tap Install when prompted.
5. 1When complete, tap Open to open the application.

10.3 FIRST-TIME SETUP

10.3.1 Registration

1. The first time you access the app, you will see the Login screen. Tap "Sign Up" to create a new account.
2. Enter your first and last name, clinic name, email address, and a password (6 characters minimum).
3. Tap "Register."
4. When asked to allow notifications, choose the option to do so. Notifications about expiration dates and insufficient stock will be sent to you via this method.
5. After completing the registration process, you will log in automatically and be taken to the dashboard.

10.3.2 Dashboard Overview

The dashboard is the main page of the app. It displays the following information about your inventory:

1. Total # of medicines - All medicine records in your inventory count.
2. Expiring Soon - The number of medicines expiring in the next 30 days.
3. Expired - The number of medicines past their expiration date.
4. Low Stock - The number of medicines in stock below their minimum threshold.
5. Total Value - The total price of all medicines in stock based on quantities combined with unit prices.
6. Quick Lists - Scrolling lists of medicines expiring soon and/or with low stock for easy access.

10.4 ADDING A MEDICINE

1. To begin adding a new medicine item, click the green plus sign icon (located as the "floating action button") located at the very bottom right corner of your Inventory screen.
2. When the new Edit Clinical Inventory item form opens, complete the following fields:
3. Medicine Name - Required: Please enter the full name of the medicine and the total dosage (if you can determine it), e.g. (Paracetamol 500mg).
4. Batch Number - Required: This field allows for the entry of the batch number/lot number located on the drug packaging and is required.
5. Current Quantity: Enter the current amount of product you have in your inventory.
6. Minimum Requirement: Enter the number of pounds of a particular product that you want to be notified via Low Stock Alert when the amount of that item reaches that point.
7. Expiration Date - Required: By clicking on the date field, you will be able to access a calendar and select the expiration date as indicated on the medicine packaging.
8. Manufactured By: Please list the manufacturer of the product in this field.
9. Category: Enter the drug category in the available category field; for example - Antibiotic, Analgesic, Antacid, Vitamin; etc.
10. Unit Cost: Each unit of the drug will have a Cost based on Wholesale Purchase Price.
11. Description: Enter any additional optional notes for your inventory as desired.
12. After completing the required data fields, click Save to create the new product on your inventory list.
13. Once new product has been created in inventory, it will now display within the inventory list with Colour Code Status based on how long after the expiration date they expire.

10.5 UNDERSTANDING STATUS BADGES

Table 10.2: Medicine Status Badge Guide

Badge Color	Status	Meaning	Action Required
Green	Safe	Expiry date is more than 30 days away	No immediate action required

Badge Color	Status	Meaning	Action Required
Amber	Expiring Soon	Expiry date is between 8 and 30 days away	Plan to use or return these medicines
Red	Critical / Expired	Expiry date is within 7 days or has already passed	Immediate action required — quarantine or dispose

10.6 MANAGING MEDICINES

10.6.1 Editing a Medicine

1. Look up the drug that you wish to modify in the Inventory List.
2. Select 'Edit' on the drug card.
3. Change what you need to in the relevant fields.
4. Press 'Save' to update the record and recalculate the status badge.

10.6.2 Deleting a Medicine

1. Look up or find the relevant drug using the Inventory List and select 'Delete' on the drug card.
2. The system will present you with a confirmation message box requesting you to confirm that you want to delete the record.
3. If you wish to remove the record permanently, select 'Delete'. If you do not want to delete the record, select 'Cancel'.

10.6.3 Searching and Filtering

1. Look up Search: Select the search bar from the top of the Inventory screen and input the name of the drug you wish to find. The list will filter real-time as you are typing.
2. Filter: Select the filter icon (top right) and choose whether you want to filter by All, Expiring Soon, Expired, or Low Stock.
3. Sort: Select the sort radio button and sort via Name (A-Z), Expiry Date (earliest first), or Quantity (lowest first).

10.7 UNDERSTANDING PUSH NOTIFICATIONS

The application will send you a push notification when there is an important inventory event that may affect your ability to manage your inventory. Push notifications will send alerts to your device when the following events occur:

- Expiry Alert - 30 Days: A medicine will expire in the next 30 days.
- Expiry Alert - 15 Days: A medicine will expire in the next 15 days. To remind you that action is needed to assist in your management of the item.
- Expiry Alert - 7 Days: A medicine will expire in the next 7 days and immediate action is required.
- Low Stock Alert: The quantity of medicine in inventory has dropped to or below the minimum amount of stock you originally specified.

Each of the notifications will indicate the specific name of the medicine and any other specific information (remaining number of days before the medicine will expire or how many total available medicine items are left in your inventory). Clicking on the push notification will take you immediately to the alerts screen to take the appropriate action. The notifications will be generated on a daily basis by the background expiry check; therefore, if your device has been powered off and then turned back on, any push notifications that were sent before the device was turned on again will automatically be generated and sent again within 60 seconds of the device being turned on.

10.8 VENDOR AND PURCHASE MANAGEMENT

10.8.1 Adding a Vendor

1. Access the Vendors section of the application either from the dashboard or from the navigation menu.
2. Press on the + to add a vendor.
3. Add the Vendor's Name, Contact Person, Phone Number, Email Address and Address.
4. Press Save. The Vendor record will now be added to your list of suppliers in the application.

10.8.2 Creating a Purchase Order

1. Go to the orders tab.
2. Select "New Order".
3. Select a vendor from your list.
4. Select medicines from your inventory.
5. Enter desired quantity and when you want it.
6. Select "Save". This saves it to your order history with a status of Pending.
7. When the order arrives, select your order record and change the status to Received.

10.9 USING THE WEBSITE CHATBOT

The project website has an AI chatbot that will help answer your application questions and direct you to common actions / tasks.

1. Go to the project website and go to the "Chatbot" page.
2. Type in the questions you have at the bottom of the screen.
3. The chatbot will return responses to your question with step-by-step answers or information relevant to your question.

Table 10.3: Sample Chatbot Questions and Responses

Sample Question	Type of Response
How do I add a medicine?	Step-by-step guide to the Add Medicine form
What does the amber badge mean?	Explanation of expiry status color coding
How do I set up a low stock alert?	Guide to setting the minimum threshold field
Why am I not receiving notifications?	Troubleshooting checklist for notification permissions
How do I add a vendor?	Step-by-step guide to the Vendor Management module
What is the expiry check frequency?	Explanation that checks run once daily via background scheduler

CHAPTER 11: SYSTEM SNAPSHOTS

We will show the main screens and user interface (UI) of the Smart Drug Inventory & Expiry Tracking System throughout this chapter, using actual screenshots from both the running Android app and website. During the development of CP-1, any new screens still under development will have been provided with a placeholder diagram until the actual screen can be inserted when that module is completed. There are numerous user interface elements that make our app successful, and we hope you enjoy using them!

11.1 ANDROID APPLICATION SCREENS

11.1.1 Splash Screen and Login Screen

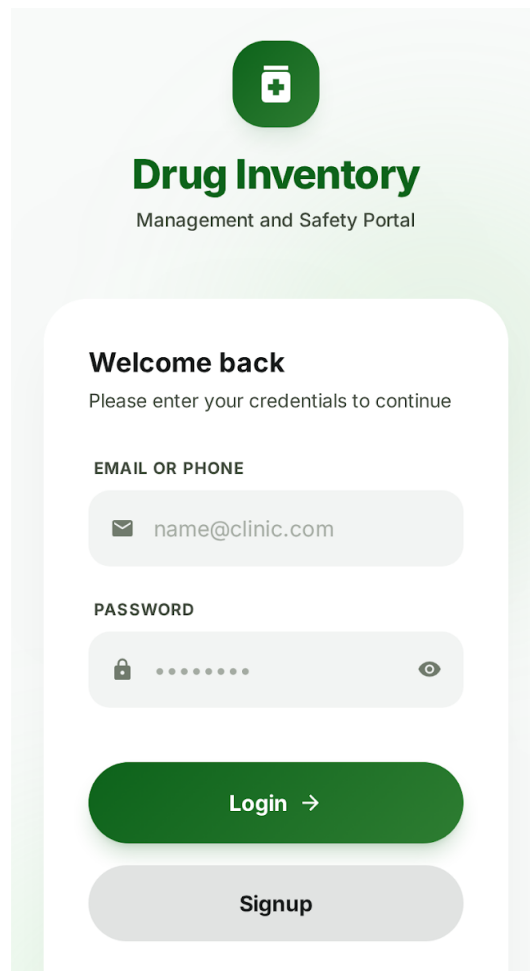


Figure 11.1: Splash Screen and Login Screen

11.1.2 Dashboard Screen

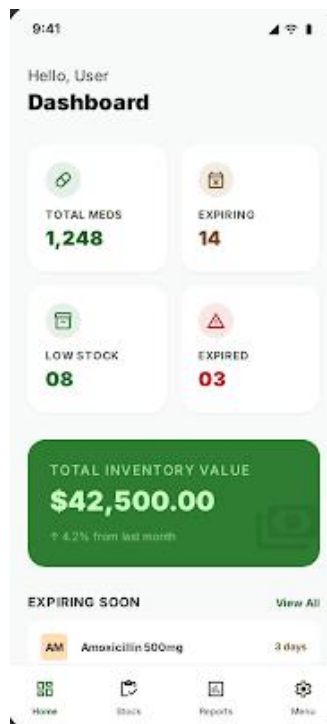


Figure 11.2: Dashboard Screen Showing Real-time Inventory Summary

11.1.3 Inventory Screen

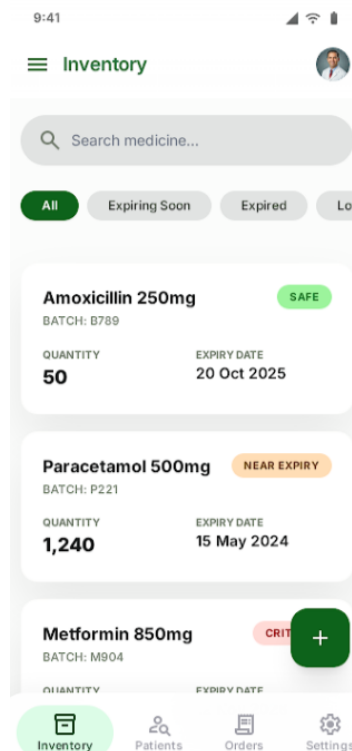
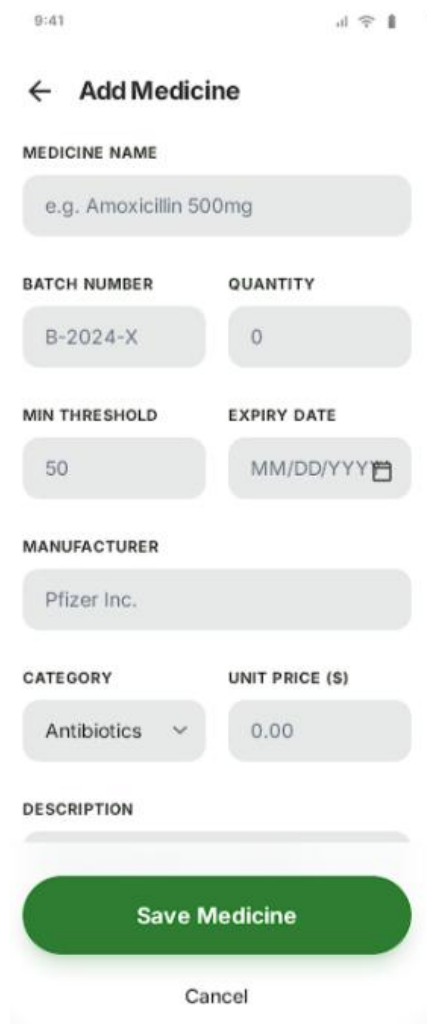


Figure 11.3: Inventory Screen with Search, Filter, and Color-coded Status Badges

11.1.4 Add Medicine Screen



A mobile application screen titled "Add Medicine". At the top, there is a back arrow and the title. Below the title, the form consists of several input fields: "MEDICINE NAME" with a placeholder "e.g. Amoxicillin 500mg", "BATCH NUMBER" with "B-2024-X", "QUANTITY" with "0", "MIN THRESHOLD" with "50", and "EXPIRY DATE" with a date picker showing "MM/DD/YYYY". Below these is a "MANUFACTURER" field with "Pfizer Inc.". Further down are "CATEGORY" (a dropdown menu showing "Antibiotics") and "UNIT PRICE (\$)" (a field with "0.00"). A "DESCRIPTION" label is followed by a large text area. At the bottom, there are two buttons: a green "Save Medicine" button and a grey "Cancel" button.

Figure 11.4: Add Medicine Form Screen

11.2 WEB PLATFORM SCREENSHOTS

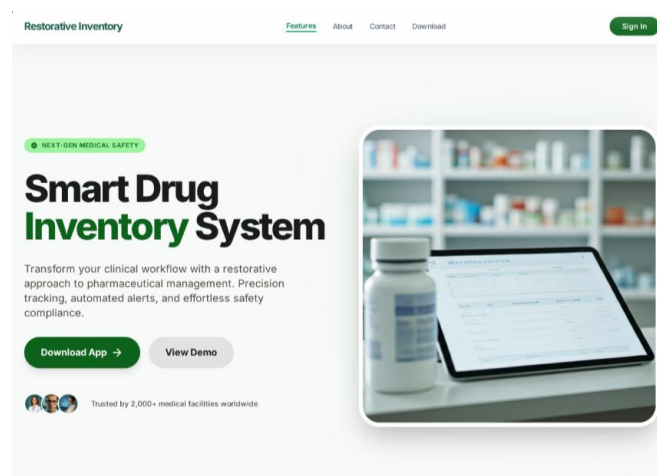


Figure 11.6: Website Landing Page (top) and AI Chatbot Interface (bottom)

11.3 PUSH NOTIFICATION EXAMPLES

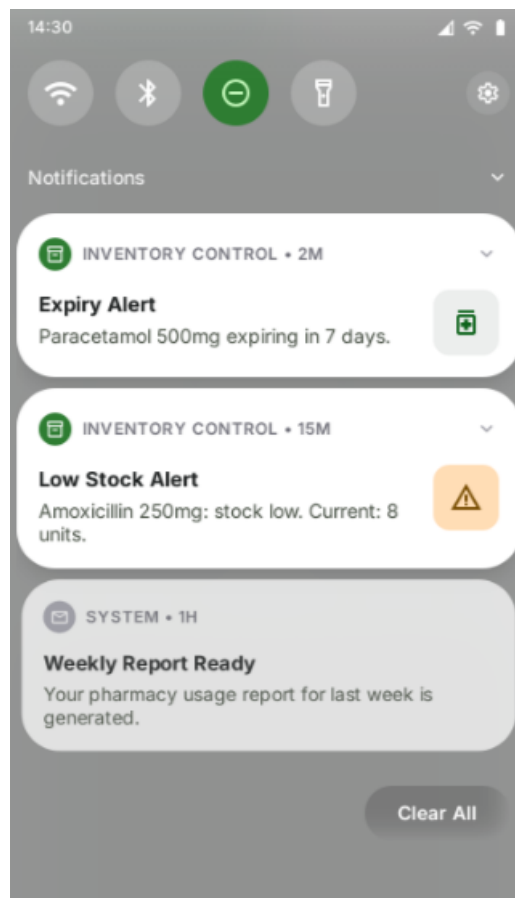


Figure 11.7: Push Notification Examples – Expiry Alert (left) and Low Stock Alert (right)

CHAPTER 12: BIBLIOGRAPHY

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